Experiment 8: Study of Common-Emitter Configuration (I_C vs V_{CE})

Arnav Kapoor (23060)

17/10/2025

Objective

To study the output characteristics of an NPN transistor in common-emitter (CE) configuration and obtain collector current (I_C) vs collector-emitter voltage (V_{CE}) plots for different base current (I_B) settings using a BC547 transistor.

Circuit Details

- Transistor: NPN BC547
- Power supply: V_{CC} (0–12 V adjustable)
- Collector resistor: variable (or multiple fixed values used to set I_C)
- Base bias network to set different base currents I_B (use series resistor and base-emitter voltage drop $V_{BE} \approx 0.7\,\mathrm{V}$)
- Instruments: digital multimeter, oscilloscope (for waveform checks), breadboard, connecting wires
- Measurement: measure I_C for a sweep of V_{CE} at several fixed base currents I_B. Record data in a table and plot I_C vs V_{CE}.

Experimental Procedure

1. Assemble the common-emitter test circuit on the breadboard. Connect collector to V_{CC} through a variable resistor or series resistors. Use a stable DC supply for V_{CC}.
2. Set a base current I_B by applying a known base resistor and measuring the base current; fix I_B for the first set of measurements.
3. For the fixed I_B, sweep V_{CE} from a low value (0 V) to the intended maximum (6–12 V) in steps (e.g., 0.5 V or 1 V). At each V_{CE}, measure and record the collector current I_C.
4. Repeat the sweep for 3–5 different base currents (I_B values) to obtain a family of I_C vs V_{CE} curves.
5. Export the recorded data to Excel and plot I_C vs V_{CE} for each I_B to identify active, saturation, and cutoff regions.
6. Optionally run LTspice simulations of the same test circuit and compare the simulation plots to experimental plots.

Observations

The observation tables and Excel plots (exported screenshots) containing the measured I_C vs V_{CE} data for different I_B values are shown below. These plots were used to identify the transistor operating regions and to extract parameters such as output resistance in the active region.

vbe	Ibe
0.1	0
0.2	0
0.3	0
0.4	0
0.5	0.05
0.6	1.75
0.7	4.53
0.8	7.65
0.9	17.5
1	26
1.2	35
1.4	46
1.6	56
1.8	73
2	88
3	203
4	480

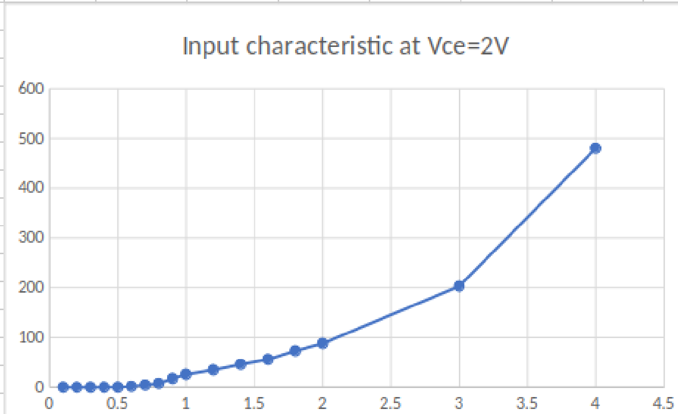


Figure 1: Excel: Measured I_C vs V_{CE} (family curves) - screenshot 1

vbe	Ibe
0.1	0
0.2	0
0.3	0
0.4	0.025
0.5	1.1
0.6	1.56
0.7	1.92
0.8	2.88
0.9	3.55
1	24
1.2	32
1.4	40
1.6	55
1.8	69
2	79
3	165
4	336

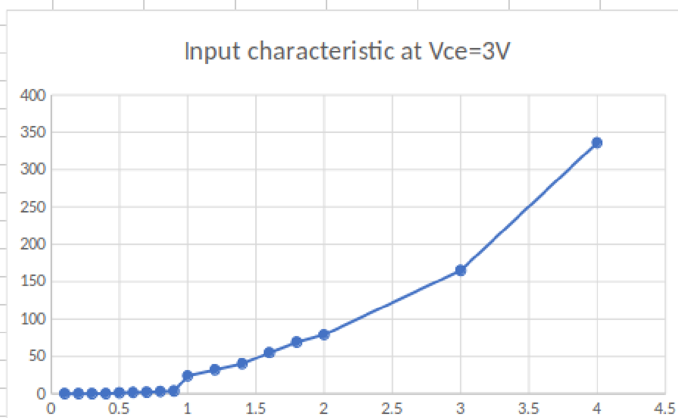
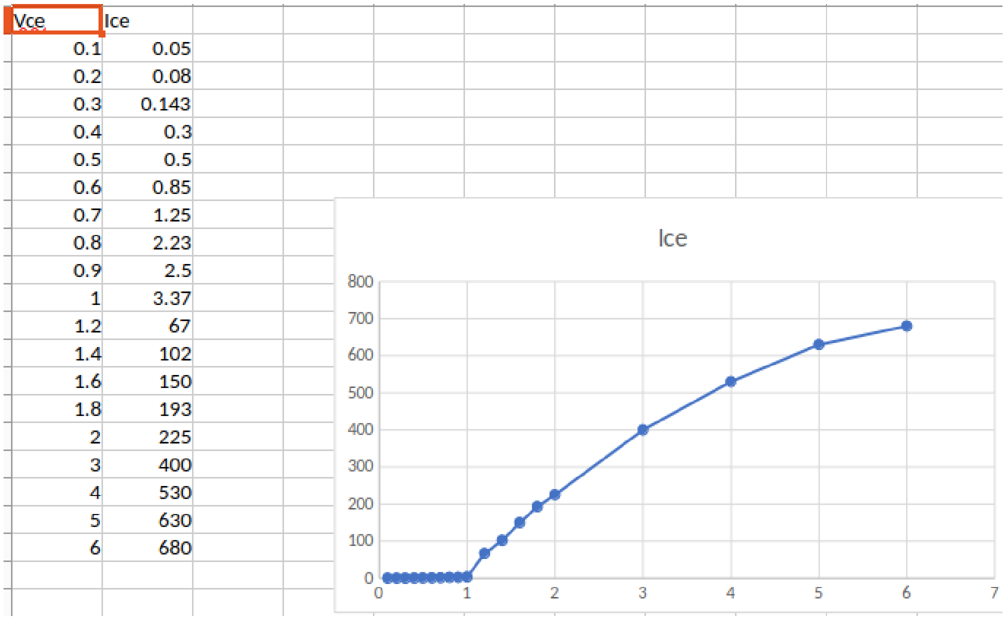
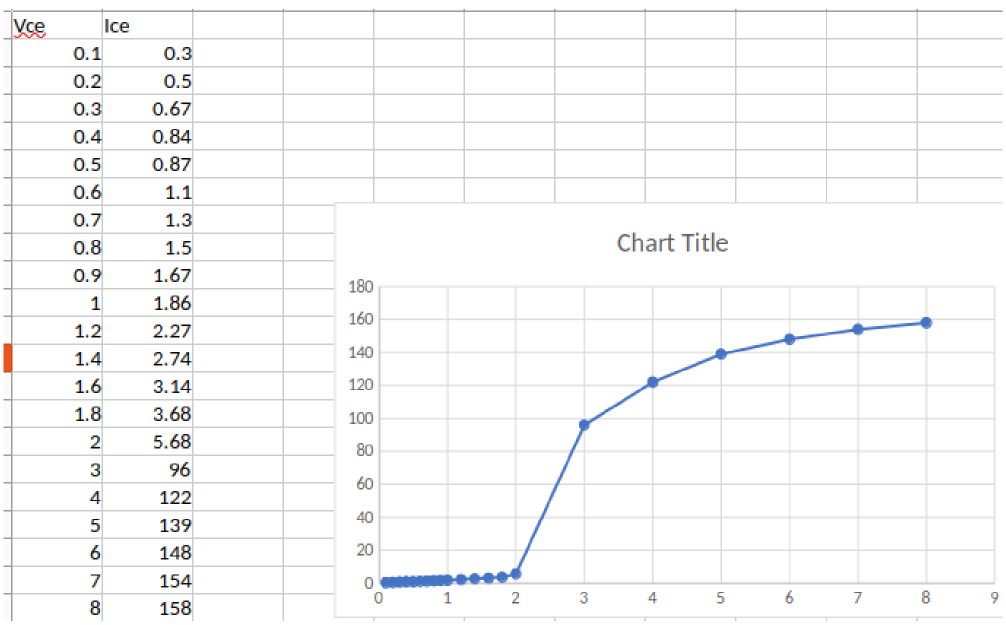


Figure 2: Excel: Measured I_C vs V_{CE} (family curves) - screenshot 2



Vce	Ice
0.1	0.05
0.2	0.08
0.3	0.143
0.4	0.3
0.5	0.5
0.6	0.85
0.7	1.25
0.8	2.23
0.9	2.5
1	3.37
1.2	67
1.4	102
1.6	150
1.8	193
2	225
3	400
4	530
5	630
6	680

Figure 3: Excel: Data table and plotted curves - screenshot 3



Vce	Ice
0.1	0.3
0.2	0.5
0.3	0.67
0.4	0.84
0.5	0.87
0.6	1.1
0.7	1.3
0.8	1.5
0.9	1.67
1	1.86
1.2	2.27
1.4	2.74
1.6	3.14
1.8	3.68
2	5.68
3	96
4	122
5	139
6	148
7	154
8	158

Figure 4: Excel: Curve fitting / slope extraction - screenshot 4

Results

From the I_C vs V_{CE} family curves obtained:

- For low V_{CE} the transistor goes into saturation (I_C limited by the external circuit); as V_{CE} increases the device enters the active region where I_C is approximately constant for a given I_B.
- The slope of I_C vs V_{CE} in the active region gives the output conductance (g_o) and its inverse the output resistance (r_o).

- The measurements allow estimation of parameters such as current gain ($\beta = I_C/I_B$) in the active region and saturation voltage $V_{CE(\text{sat})}$ when the device is fully driven.

LTspice Setup

Photos of the common-emitter test setup (connections, supply, and measurement points):

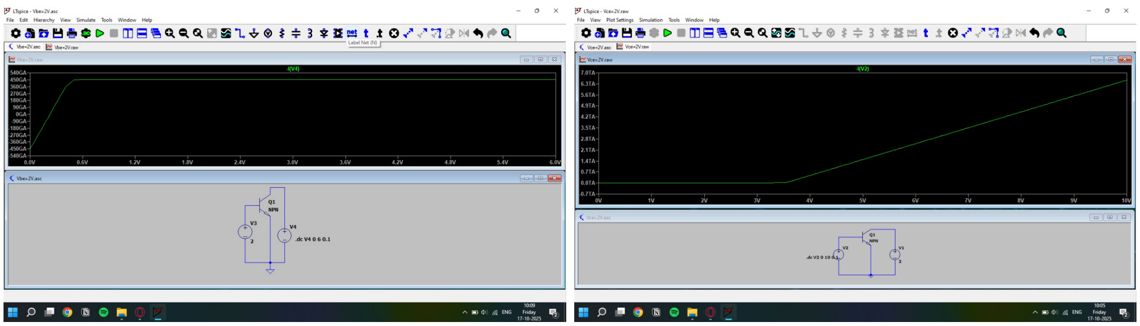


Figure 5: Breadboard wiring and measurement using multimeter/ammeter.

Discussion

The experimental I_C vs V_{CE} curves show the expected behavior for a transistor in CE configuration: cutoff at very low base currents, an active region where I_C is primarily controlled by I_B, and saturation at low V_{CE}. Differences between the experimental and ideal/simulated plots can be attributed to:

- Component tolerances (resistor tolerances, actual transistor β variation between samples)
- Measurement errors (meter/ammeter burden voltage, probe loading)
- Temperature effects during measurement which change device characteristics
- Wiring resistance and contact resistance on the breadboard

Conclusion

The common-emitter experiment (Objective 7.1) successfully produced a family of I_C vs V_{CE} curves for the BC547 transistor. The active, saturation, and cutoff regions were observed and characterized. Measured current gain values and output resistance estimates are consistent with typical BC547 datasheet ranges, within experimental uncertainty. Comparison with spreadsheet plots confirms the trends and provides a basis for further quantitative analysis (e.g., extracting r_o and β).